

Wireshark interface showing HTTP traffic. The packet list on the left shows a GET request to /portal_css/01dn20P1onek203n20Custom20Theme/at-cachekey2149.kss. The packet details pane on the right shows the request structure, including the URL, headers, and body. The packet bytes pane at the bottom shows the raw data.

Wireshark interface showing DNS traffic. The packet list on the left shows a standard query for 0xc201 A fonts.googleapis.com. The packet details pane on the right shows the query structure, including the question, answer, and authority sections. The packet bytes pane at the bottom shows the raw data.

Wireshark interface showing TCP traffic. The packet list on the left shows a SYN packet from 192.168.52.8 to 142.250.67.202. The packet details pane on the right shows the SYN packet structure, including the source and destination ports, sequence number, and window size. The packet bytes pane at the bottom shows the raw data.

Nmap fundamentals

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Session Actions Edit View Help
blackking@kali -

(blackking@kali)-[~]
$ nmap 192.168.52.7
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-08 10:01 +0530
Nmap scan report for 192.168.52.7
Host is up (0.0022s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
445/tcp   open  microsoft-ds
631/tcp   open  ipp
3000/tcp  closed ppp
3306/tcp  open  mysql
8080/tcp  closed http-proxy
8181/tcp  closed intermapper
MAC Address: 08:00:27:F7:07:13 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 6.17 seconds

(blackking@kali)-[~]
$
```

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Session Actions Edit View Help
blackking@kali -

SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES
Scantype v not supported

(blackking@kali)-[~]
$ nmap -O 192.168.52.7
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-08 10:04 +0530
Nmap scan report for 192.168.52.7
Host is up (0.0033s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
445/tcp   open  microsoft-ds
631/tcp   open  ipp
3000/tcp  closed ppp
3306/tcp  open  mysql
8080/tcp  open  http-proxy
8181/tcp  closed intermapper
MAC Address: 08:00:27:F7:07:13 (Oracle VirtualBox virtual NIC)
Aggressive OS guesses: Linux 3.2 - 4.14 (98%), Linux 3.8 - 3.16 (98%), Linux 3.10 - 4.11 (94%), Linux 3.13 - 4.4 (94%), Linux 3.13 (94%), Linux 3
.13 - 3.16 (94%), OpenWrt Chaos Calmer 15.05 (Linux 3.18) or Designated Driver (Linux 4.1 or 4.4) (94%), Linux 4.10 (94%), Android 5.0 - 6.0.1 (L
inux 3.4) (94%), Android 8 - 9 (Linux 3.18 - 4.4) (94%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.48 seconds

(blackking@kali)-[~]
$
```

```

blackking@kali: ~
$ nmap -sn 192.168.52.7
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-08 10:08 +0530
Nmap scan report for 192.168.52.7
Host is up (0.0017s latency).
MAC Address: 08:00:27:F7:07:13 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.63 seconds

blackking@kali: ~
$ nmap -p 192.168.52.7
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-08 10:09 +0530
Error #487: Your port specifications are illegal. Example of proper form: "-100,200-1024,T:3000-4000,U:60000-"
QUITTING!

blackking@kali: ~
$ nmap -p 80 192.168.52.7
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-08 10:09 +0530
Nmap scan report for 192.168.52.7
Host is up (0.0036s latency).

PORT      STATE SERVICE
80/tcp    open  http
MAC Address: 08:00:27:F7:07:13 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.75 seconds

blackking@kali: ~
$ nmap -sV 192.168.52.7
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-08 10:09 +0530
Nmap scan report for 192.168.52.7
Host is up (0.0021s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION

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```

blackking@kali: ~
$ nmap -A 192.168.52.7
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-08 10:11 +0530
Nmap scan report for 192.168.52.7
Host is up (0.0031s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          ProFTPD 1.3.5
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 2b:2e:1f:a4:54:26:87:76:12:26:59:58:0d:da:3b:04 (DSA)
|   2048 c9:ac:70:ef:f8:de:8b:a3:a3:44:ab:3d:32:0a:5c:6a (RSA)
|   256  c0:49:cc:18:7b:27:a4:07:0d:2a:0d:bb:42:4c:36:17 (ECDSA)
|_  256  a0:76:f3:76:f8:f0:70:4d:09:ca:e1:10:fd:a9:cc:0a (ED25519)
80/tcp    open  http         Apache httpd 2.4.7
|_ http-ls: Volume /
|_  SIZE  TIME      FILENAME
|_  -    2020-10-29 19:37  chat/
|_  -    2011-07-27 20:17  drupal/
|_  1.7K  2020-10-29 19:37  payroll_app.php
|_  -    2013-04-08 12:06  phpmyadmin/
|_
|_ http-title: Index of /
|_ http-server-header: Apache/2.4.7 (Ubuntu)
445/tcp   open  netbios-ssn  Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
631/tcp   open  ipp          CUPS 1.7
|_ http-title: Home - CUPS 1.7.2
|_ http-robots.txt: 1 disallowed entry
|_ /
|_ http-server-header: CUPS/1.7 IPP/2.1
|_ http-methods:
|_  Potentially risky methods: PUT
3000/tcp  closed ppp

```

Wireshark interface showing a packet capture on the 'eth0' interface. The packet list on the left shows a series of UDP packets from 192.168.52.8 to 192.168.52.8. The selected packet (No. 704) is a UDP packet with source 192.168.52.8 and destination 192.168.52.8. The packet details pane shows the User Datagram Protocol (UDP) section, indicating a source port of 443 and a destination port of 37375. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header.

Wireshark interface showing a packet capture on the 'eth0' interface. The packet list on the left shows a series of TCP packets from 192.168.52.8 to 192.168.52.8. The selected packet (No. 592) is a TCP packet with source 192.168.52.8 and destination 192.168.52.8. The packet details pane shows the Transmission Control Protocol (TCP) section, indicating a source port of 443 and a destination port of 37375. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and Transmission Control Protocol header.

Wireshark interface showing a packet capture on the 'eth0' interface. The packet list on the left shows a series of IP packets from 192.168.52.8 to 192.168.52.8. The selected packet (No. 689) is an IP packet with source 192.168.52.8 and destination 192.168.52.8. The packet details pane shows the Internet Protocol Version 4 section, indicating a source address of 192.168.52.8 and a destination address of 192.168.52.8. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header.